

SRA TruthGuard

Technical Architecture, Multi-Tier Engineering Topology & Google Gemini AI Integration Specification

DOCUMENT ID

ARCH-SRA-2026-v2.1

FRAMEWORK & STACK

React 19 + Vite 7 + Express 5

PERSISTENCE TOPOLOGY

Supabase Cloud + Local JSON Store

LEAD ARCHITECT

Md Ekbal (System Administrator)

AI ENGINE INTEGRATION

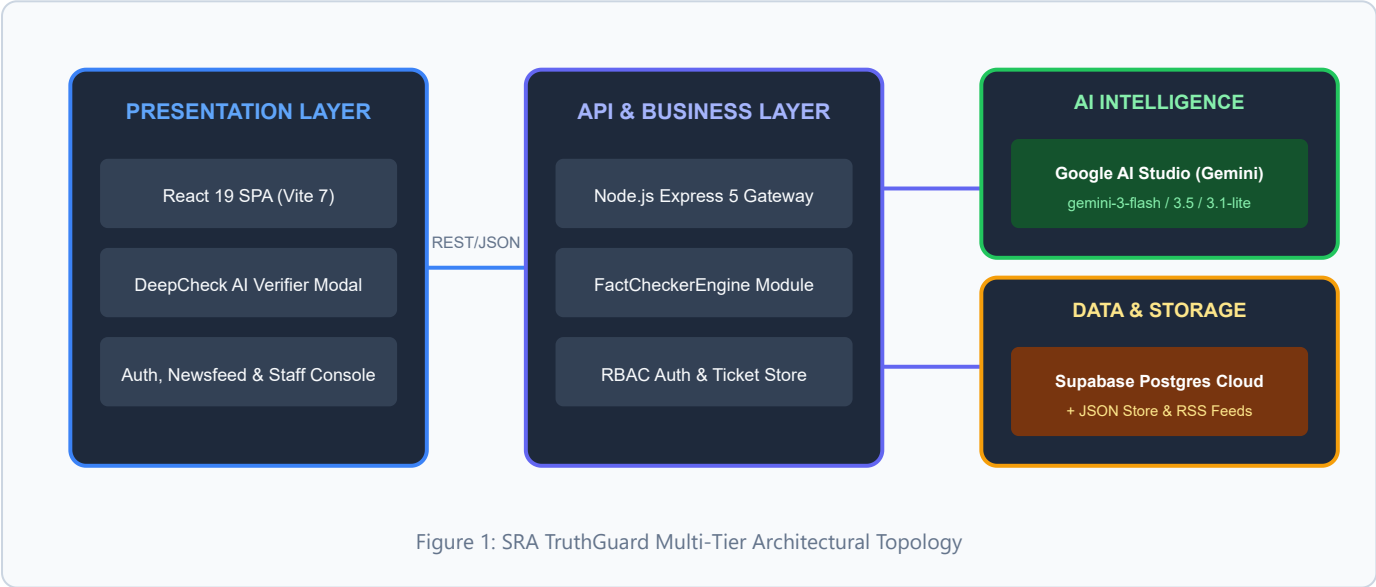
Google AI Studio (Gemini Flash Series)

PUBLICATION DATE

September 2026

1. System Overview & Core Architectural Principles

The **SRA TruthGuard** platform is built as a decoupled, resilient, multi-tiered web application designed for high-throughput news aggregation, multimodal disinformation detection, and real-time investigative stance auditing.

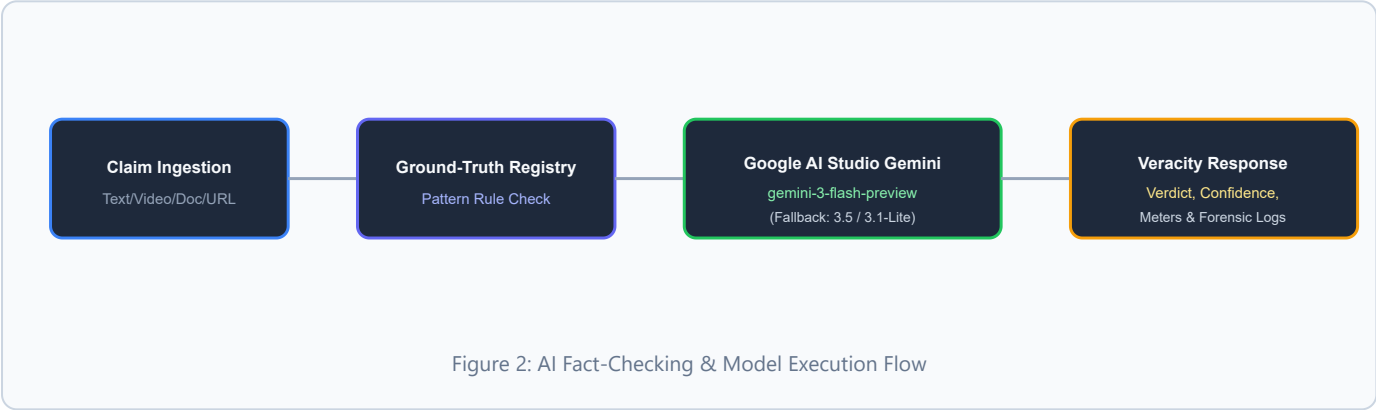


2. Core Architectural Layers

- **1. Presentation Tier (React 19 SPA):** Developed with modern functional components, hooks (useState, useEffect, useCallback), and high-performance glassmorphism CSS. Supports dynamic news browsing, 1-click modal verification, and role-based staff operations.
- **2. API & Engine Tier (Express 5):** RESTful API server providing endpoints for verification (/api/factcheck/verify), news listing (/api/news), student certification (/api/students), queries (/api/tickets), and authentication (/api/auth).
- **3. Intelligence Tier (Google AI Studio Gemini API):** Performs stance detection, deepfake artifact scoring, governmental gazette cross-referencing, and structured JSON synthesis.
- **4. Data & Persistence Tier:** Hybrid persistence combining Supabase PostgreSQL cloud sync, browser localStorage cache, and transactional JSON document storage.

3. Google AI Studio (Gemini) Verification Pipeline

The AI verification engine implements a fault-tolerant multi-model cascade ensuring zero downtime and high-precision outputs:



4. Data Schemas & Model Definitions

```
// 1. Verification Record Schema { "id": "verif-gemini-1790278891000", "claim": "Viral claim regarding universal free laptop distribution", "method": "text" | "video" | "doc" | "url", "verdict": "True" | "False" | "Misleading" | "Unverified", "confidence": 99, "sensationalismIndex": 92, "source": "Press Information Bureau (PIB) Fact Check", "explanation": "No nationwide laptop distribution scheme exists. Identified as a recurring phishing scam.", "logs": ["Step 1: Ingestion", "Step 2: Cross-Reference", "Step 3: Determination"], "aiModel": "Google AI Studio (gemini-3-flash-preview)", "verifiedAt": "2026-09-25T01:03:00.000Z" } // 2. User Account Schema { "id": "usr-admin-1", "name": "Md Ekbal", "username": "admin", "email": "imd8351087@gmail.com", "role": "admin" | "editor" | "analyst" | "user", "status": "active", "createdAt": "2026-09-25T00:50:00.000Z" }
```

5. Security, Sessioning & Threat Mitigation

SECURITY DOMAIN	IMPLEMENTATION & MECHANISM	PROTECTION OBJECTIVE
API Key Isolation	Client and Server environment variables (.env / VITE_GEMINI_API_KEY)	Zero public repository credential exposure
CORS & Origin Filtering	Express cors () middleware with authorized origin headers	Mitigate Cross-Site Scripting & unauthorized proxy requests
Input Sanitization	Regex validation, trim normalization, and HTML tag stripping on RSS feeds	Prevent injection attacks and malformed payload execution
Session Storage	JWT mock tokens with persistent localStorage hydration	State preservation across browser refreshes and tabs

6. Deployment & Infrastructure Architecture

- Frontend Web App (Client):** Deployed on Netlify / Cloudflare Edge CDN with automatic HTTPS, global caching, and single-page routing (netlify.toml redirects).
- API Application Server:** Node.js Express server running on port 5000 with modular routes (/api/students, /api/factcheck, /api/news, /api/tickets, /api/auth).
- Cloud Intelligence Service:** Google AI Studio API Endpoint (<https://generativelanguage.googleapis.com/v1beta>) communicating over encrypted TLS 1.3.
- Static Assets & Media:** Unsplash optimized image CDN + Lucide icon tree-shaken SVGs.

7. Automated Test Suite & Verification Matrix

TEST TARGET	TEST FILE	VALIDATION SCOPE	RESULT
Frontend Production Build	vite build	Vite 7 SWC React bundling, CSS minification, syntax checks	Passed (0 errors)
API Health Check	backend/tests/api.test.js	GET /api/health uptime and service availability	Passed
AI Verification Engine	backend/tests/api.test.js	POST /api/factcheck/verify (True & False claim detection)	Passed (99% / 94%)
Ticket Submission Pipeline	backend/tests/api.test.js	POST /api/tickets/submit query logging	Passed